

MATHIAS VERNIER

EMBEDDED SYSTEMS ENGINEER

EDUCATION

ECE Paris - Graduate School of Engineering 2023 – 2028

Master's Degree in Engineering – Major in Embedded Systems (ING4)

- Seeking a 5-month International Engineering Internship (April – August 2027).
- **Relevant Coursework:** Real-Time OS, Computer Architecture, Signal Processing, Hardware AI, Control Theory, Embedded Systems Design, Digital Logic & RTL.

EXPERIENCE

Engineering Intern – IMOCA Digital Twin Jan 2026 – Feb 2026

4CAD Group – Vendée Globe 2028 Project

- Optimized native CAD data to build a real-time 3D VR cockpit validation tool (60+ FPS) for skipper Benjamin Dutreux.
- Engineered a headset-free cockpit walkthrough using motion tracking sensors, deployed live at Global Industrie 2026.
- Created sensor-driven virtual environments and photorealistic assets with KeyShot for pre-manufacturing reviews.

Engineering Intern – Automated Workflows Jan 2025 – Feb 2025

Groupe SYD

- Architected end-to-end automated data workflows, reducing operational error rates and improving data integrity.
- Enhanced UI/UX interfaces through rapid user feedback iterations to maximize cross-team software adoption.
- Managed complete product lifecycle: requirements analysis, software logic implementation, testing, and deployment.

TECHNICAL PROJECTS

Fruit Ninja on FPGA – Pure Hardware RTL Academic Project

Intel MAX 10 (DE10-Lite) · VHDL · Quartus Prime · VGA

- Engineered an arcade game in pure RTL VHDL (no processor, zero OS overhead) on dedicated 25 MHz synchronous pipelines.
- Designed a custom VGA controller (640×480 @ 60 Hz, 12-bit DAC) and contactless control via HC-SR04 ultrasonic sensor.
- Simulated real-time parabolic physics for 13 concurrent sprites with fixed-point tangent LUTs for slicing collisions.

Drawbot – Autonomous Plotter Robot Academic Project

ESP32 · C++ · PID Control · Kinematics · Wi-Fi

- Implemented dual closed-loop PID control (linear and angular) on ESP32 with encoder interrupts for trajectory tracking.
- Derived differential kinematics with 13 cm pen-offset compensation to draw complex shapes (rosettes, circles).
- Embedded an asynchronous Wi-Fi web server (ESP32 AP) for remote control; debugged timers via logic analyzer & scope.

SUMMARY

ING4 (Master 1) Embedded Systems Engineering student at ECE Paris specializing in digital hardware logic (pure RTL VHDL) and real-time firmware control. Proven experience developing industrial VR digital twins (4CAD Group / Vendée Globe 2028) and autonomous kinematic robots with dual PID loops. Seeking a 5-month international engineering internship (April – August 2027) to contribute to cutting-edge embedded systems.

CONTACT

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[mathiassece.github.io](https://github.com/mathiassece)

TECHNICAL SKILLS

Programming

C, C++, VHDL, Python, Java, JavaScript

Hardware & Embedded

Intel MAX 10, ESP32, STM32, ARM Cortex, RTOS, PID Control

Tools & Lab Equipment

Logic Analyzer, Oscilloscope, Quartus Prime, Git, Scilab

Simulation & Graphics

VR Simulation, Motion Sensing, KeyShot, CAD, 3D Rendering

LANGUAGES

French: Native

English: Advanced (TOEIC 870)

EXTRACURRICULAR

Paris Fire Brigade (BSPP)

First responder in full-scale counter-terrorism & mass casualty crisis drill.